

**Design and Implementation of a Smart Stethoscope using STM32 and
Cube Monitor**

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ABSTRACT

This project presents the **design and development of a Smart Stethoscope** — an embedded system capable of **digitally acquiring, processing, and analyzing heart sounds** in real time using an **STM32 Nucleo-L152RE microcontroller**. The system employs a **diaphragm-based condenser microphone** that captures the low-frequency acoustic vibrations generated by the heart during cardiac cycles. These analog signals are conditioned, amplified, and then digitized through the **12-bit ADC (Analog-to-Digital Converter)** of the STM32 microcontroller.

The data acquisition process is synchronized using **TIM2 timer triggering** and **Direct Memory Access (DMA)**, ensuring consistent and jitter-free sampling at a rate of **1 kHz** without burdening the CPU. The digitized samples are transmitted via a **Bluetooth HC-05 module** to a host system, where the **CubeMonitor** software (powered by Node-RED) performs real-time signal visualization and analysis.

The heart sound signal undergoes a sequence of digital signal processing stages implemented using Node-RED function nodes. First, an **Exponential Moving Average (EMA)** algorithm removes the DC bias and stabilizes the baseline drift. Next, a **band-pass filter (20–200 Hz)** isolates the primary heart sound components corresponding to **S1 (lub)** and **S2 (dub)** events, effectively suppressing ambient noise and respiratory interference. Subsequently, **envelope detection** is applied to extract the amplitude profile of the heartbeat waveform. This envelope signal is then analyzed to identify periodic peaks, from which the **Beats Per Minute (BPM)** is computed, providing an accurate estimation of the heart rate.

The proposed Smart Stethoscope system offers a **low-cost, portable, and power-efficient solution** for non-invasive cardiac monitoring. Its modular design

enables easy integration with other biomedical sensors and cloud-based platforms for telemedicine applications. With further enhancements, such as feature extraction and machine learning-based heart sound classification, this prototype can evolve into a **comprehensive diagnostic assistant** capable of detecting cardiac abnormalities remotely.

INTRODUCTION

Cardiovascular diseases remain one of the leading causes of mortality worldwide, highlighting the need for accessible and continuous cardiac monitoring solutions. Conventional stethoscopes, though widely used, rely entirely on the clinical expertise of the practitioner and offer no means of recording, storing, or analyzing heart sounds. With advancements in embedded systems, low-power microcontrollers, and digital signal processing techniques, **electronic or “smart” stethoscopes** have emerged as an important tool for objective and remote cardiac assessment.

Heart sounds, primarily the **S1 (lub)** and **S2 (dub)** components, contain essential diagnostic information related to valve movements, cardiac rhythm, and overall heart health. However, these signals are typically low in amplitude and lie within the **20–200 Hz frequency range**, making them susceptible to noise, motion artefacts, and environmental disturbances. Capturing and analyzing these subtle vibrations requires a sampling system with stable timing, proper signal conditioning, and robust digital filtering.

This project focuses on building a **Smart Stethoscope system** using the **STM32 Nucleo-L152RE**, a low-power ARM Cortex-M3 microcontroller capable of precise analog data acquisition. A **diaphragm-based microphone** is used as the sensing element to convert acoustic heart vibrations into analog voltage signals. These signals are digitized using the microcontroller’s **12-bit ADC** with a stable sampling rate generated through **TIM2 hardware triggering** and **DMA-based data transfer**. This architecture ensures consistent timing, minimal CPU overhead, and high-quality signal capture suitable for biomedical analysis.

For processing and visualization, the digitized samples are transmitted via a **Bluetooth HC-05 module** to a computer where **CubeMonitor (Node-RED)**

performs real-time waveform display and analysis. A carefully designed DSP pipeline is implemented using Node-RED function nodes, including **bias removal, IIR band-pass filtering, envelope extraction, and heartbeat peak detection**. These processing stages enhance the visibility of S1/S2 components and enable accurate computation of the **Beats Per Minute (BPM)**, which serves as the primary physiological metric for assessing cardiac activity.

The aim of this work is to demonstrate a **cost-effective, portable, and user-friendly digital stethoscope** capable of supporting basic cardiac monitoring tasks. The system can assist medical practitioners in screening for abnormalities such as tachycardia, bradycardia, irregular rhythms, and abnormal heart sound patterns. With further expansion into machine learning-based murmur detection or cloud-based patient monitoring, this Smart Stethoscope has the potential to support scalable telemedicine, especially in resource-limited environments.

METHODOLOGY

The methodology of the Smart Stethoscope system consists of five major stages:

1. Signal Acquisition

A diaphragm-based electret microphone is used to capture mechanical vibrations produced by the heart. These vibrations are converted into an electrical signal and fed into the **ADC input (PA0)** of the **STM32 Nucleo-L152RE** board.

Key considerations:

- Heart sounds lie in **20–200 Hz**
- Microphone output is weak and requires careful sampling
- Environmental noise and motion artefacts must be filtered out

The STM32's 12-bit ADC samples the signal with a stable **1 kHz sampling rate**, generated using **TIM2 hardware trigger**. This ensures:

- No jitter
- Uniform sampling

- High-quality biomedical data

2. ADC Sampling Using DMA and TIM2

To achieve continuous, lossless sampling, the following configuration is used:

TIM2 generates a trigger at 1 kHz

- PSC = 31
- ARR = 499
- → Update event every 1 ms

ADC operates in DMA mode

- DMA fills a 512-sample buffer
- Two callbacks used:
 - HAL_ADC_ConvHalfCpltCallback()
 - HAL_ADC_ConvCpltCallback()

This allows:

- Zero CPU involvement during sampling
- No missed samples
- Smooth and reliable data flow

3. Data Transmission via Bluetooth HC-05

Every 8th sample is transmitted at **9600 baud** to prevent buffer overflow.

- 9600 baud $\approx$ 960 bytes/sec
- Each sample $\approx$ 6 bytes
- 125 samples/sec $\leftarrow$ safe, lossless transmission

The STM32 sends data over:

- **USART1 → HC-05 → PC**

Tera Term verifies transmission, and finally CubeMonitor receives the stream.

4. Digital Signal Processing (CubeMonitor / Node-RED)

DSP is implemented using sequential Function nodes:

a. Bias Removal (EMA)

Removes DC offset from the mic signal using Exponential Moving Average (EMA):

- Centers waveform around zero
- Removes slow drift
- Prepares signal for filtering

b. Band-Pass IIR Filtering (20–200 Hz)

A biquad IIR filter is applied:

- Removes high-frequency noise
- Suppresses low-frequency rumble
- Retains only heart-sound components (S1/S2)

c. Envelope Detection

- xHigh-pass filter (for drift removal)
- Absolute value
- Exponential smoothing

This produces a clean amplitude envelope which highlights the S1/S2 peaks.

d. BPM Calculation

Using envelope peaks:

- Detect threshold crossings
- Measure time interval between detected beats

- Compute BPM = 60000 / interval

The BPM is displayed on a Gauge widget in CubeMonitor.

5. Real-Time Visualization Dashboard

CubeMonitor dashboard displays:

- **Raw signal waveform**
- **Filtered signal waveform**
- **Envelope signal**
- **Real-time BPM**
- **BPM classification (Low/Normal/High)**

This allows continuous monitoring and interpretation of heart sound data.

RESULTS

The system successfully captured and processed heart sound signals using the STM32 Nucleo-L152RE. The observed outputs include:

1. Raw Heartbeat Waveform

- Centered around 2000–2200 ADC counts
- Visible spikes corresponding to microphone vibration
- Some noise due to environment (expected)

2. Filtered (20–200 Hz) Heart-Sound Waveform

- S1 and S2 heart sounds clearly highlighted
- Noise and low-frequency drift suppressed
- Waveform smooth and suitable for envelope extraction

3. Envelope Signal

- Smooth rising peaks
- Each strong peak corresponds to one heartbeat
- Consistent amplitude when diaphragm is placed properly

4. BPM Measurement

Average BPM range observed: **70–90 BPM** (depending on user & conditions)

Matches physiological norms:

- Normal resting: 60–100 BPM

5. Dashboard Performance

- Continuous scrolling window of last 10 seconds
- Clean real-time graphs
- BPM gauge updates every heartbeat
- Classification text node shows:
 - **Low Heart Rate (<60 BPM)**
 - **Normal (60–100 BPM)**
 - **High (>100 BPM)**

INFERENCE

From the experimental results, the following inferences are made:

1. **Stable Sampling Achieved** TIM2 + DMA ensures jitter-free sampling suitable for biomedical applications.
2. **Effective Noise Reduction** EMA bias removal and IIR band-pass filtering significantly reduce environmental noise.
3. **Accurate Beat Detection** Envelope detection successfully isolates S1/S2 peaks.
4. **Reliable BPM Estimation** Computed BPM matches real physiological values.

5. **Usability** Real-time visualization and Bluetooth transmission create a user-friendly monitoring interface.
6. **Potential for Diagnostics** Although this version primarily estimates BPM, the clean S1/S2 signal can be extended to detect abnormalities such as valve issues, murmurs, or arrhythmias.

FUTURE SCOPE

The Smart Stethoscope can be extended into a more advanced diagnostic tool. Possible future enhancements include:

1. ML-Based Murmur Detection

- Train ML models (CNN, SVM) to classify:
 - Murmur / No murmur
 - Systolic vs Diastolic issues
 - Abnormal vs Normal heart sounds

2. Cloud-Connected Health Monitoring

- Upload signal + BPM to cloud (Firebase / AWS)
- Doctor can monitor remotely

3. Improved Hardware

- Add analog front-end (AFE) for amplification & filtering
- Use MEMS microphones for higher signal clarity

4. Advanced DSP Metrics

- S1/S2 ratio
- Heart sound intensity
- Timing analysis (S1–S2 duration)
- Signal quality index

5. Mobile App

- Real-time Bluetooth visualizer on Android (Flutter)

6. Multi-Channel Stethoscope

- Compare two chest locations simultaneously
- Useful for murmur localization

REFERENCES

Here are some strong references relevant to your work:

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